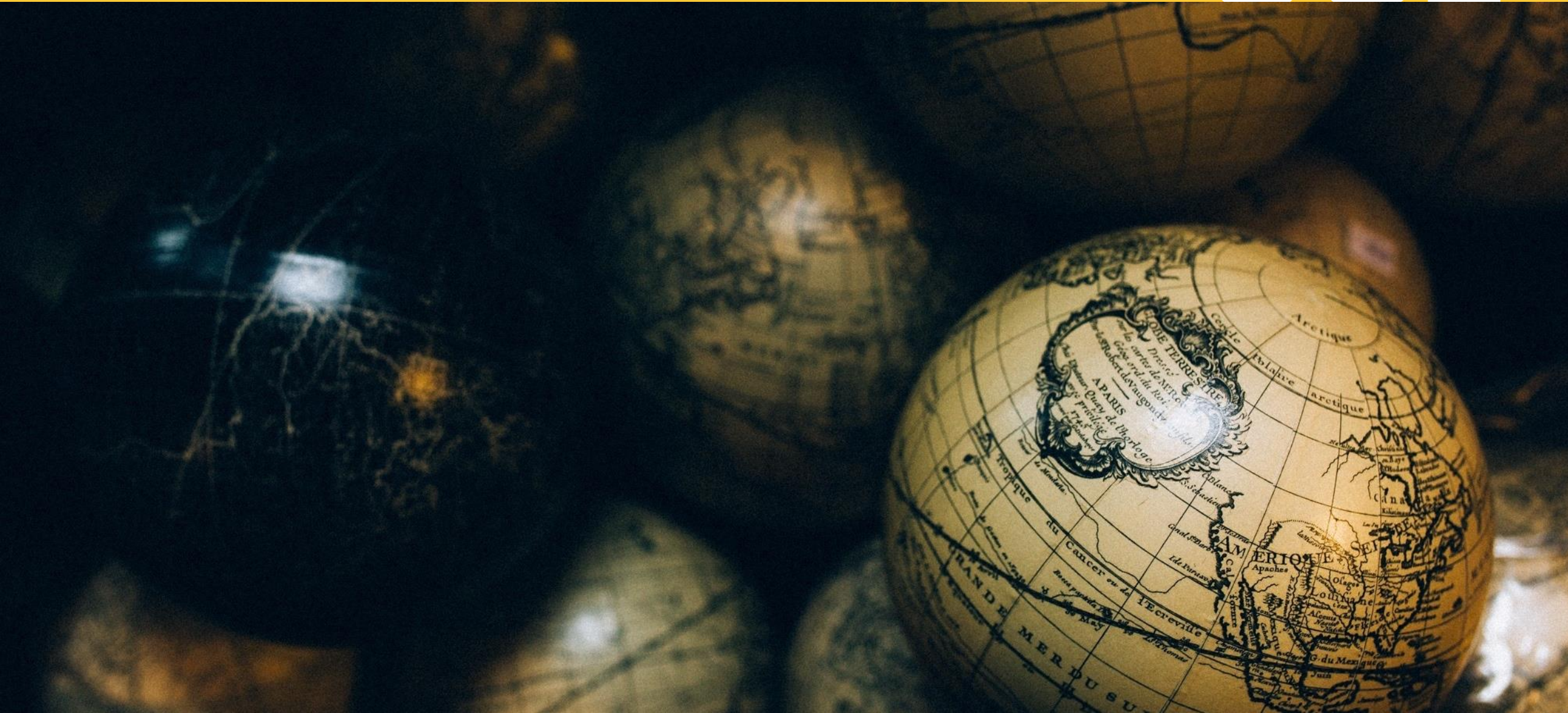


Situating the Atlas



UCL

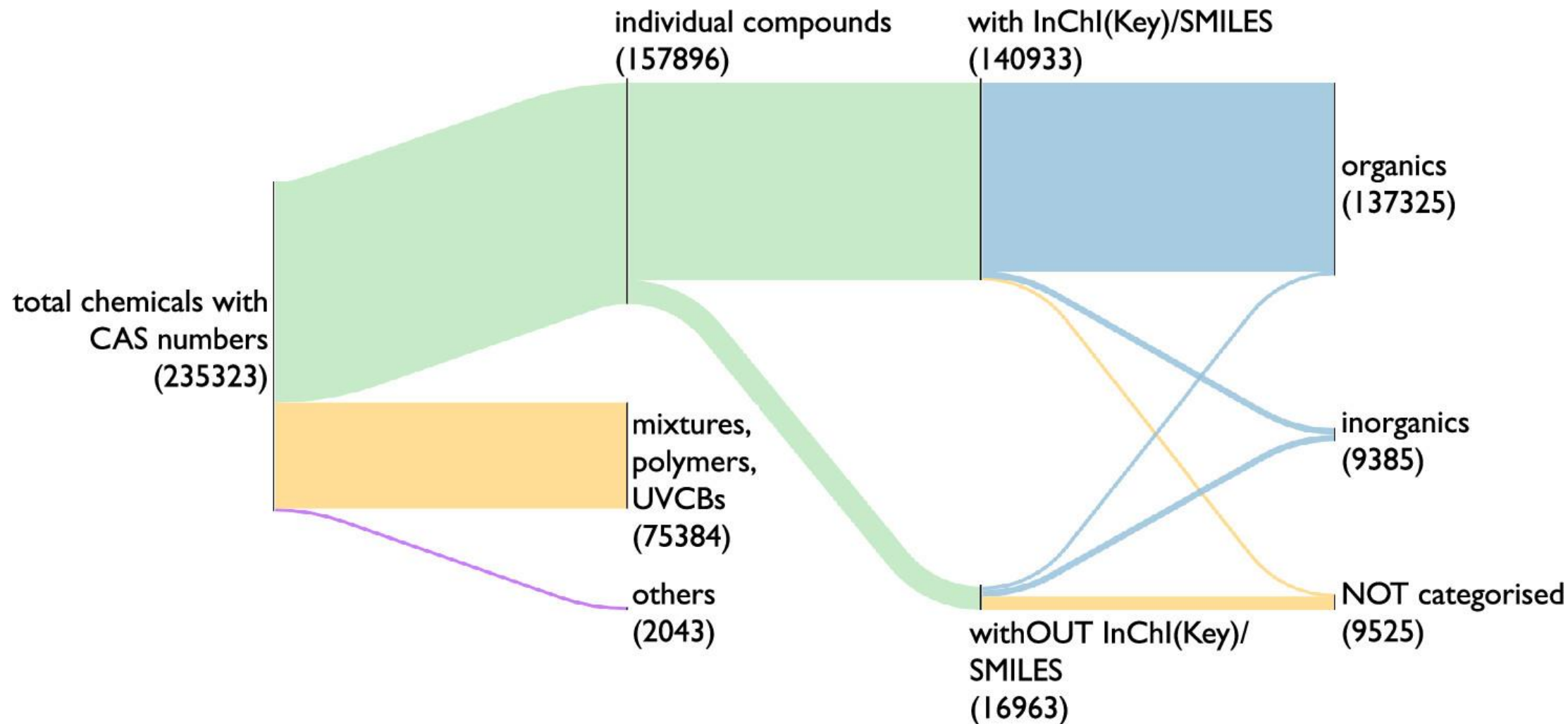


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Toward a Global Understanding of Chemical Pollution: A First Comprehensive Analysis of National and Regional Chemical Inventories

Zhanyun Wang, Glen W. Walker, Derek C. G. Muir, and Kakuko Nagatani-Yoshida

Environmental Science & Technology **2020** 54 (5), 2575-2584, DOI: 10.1021/acs.est.9b06379



Plastic Chemicals: <https://plastchem-project.org/>

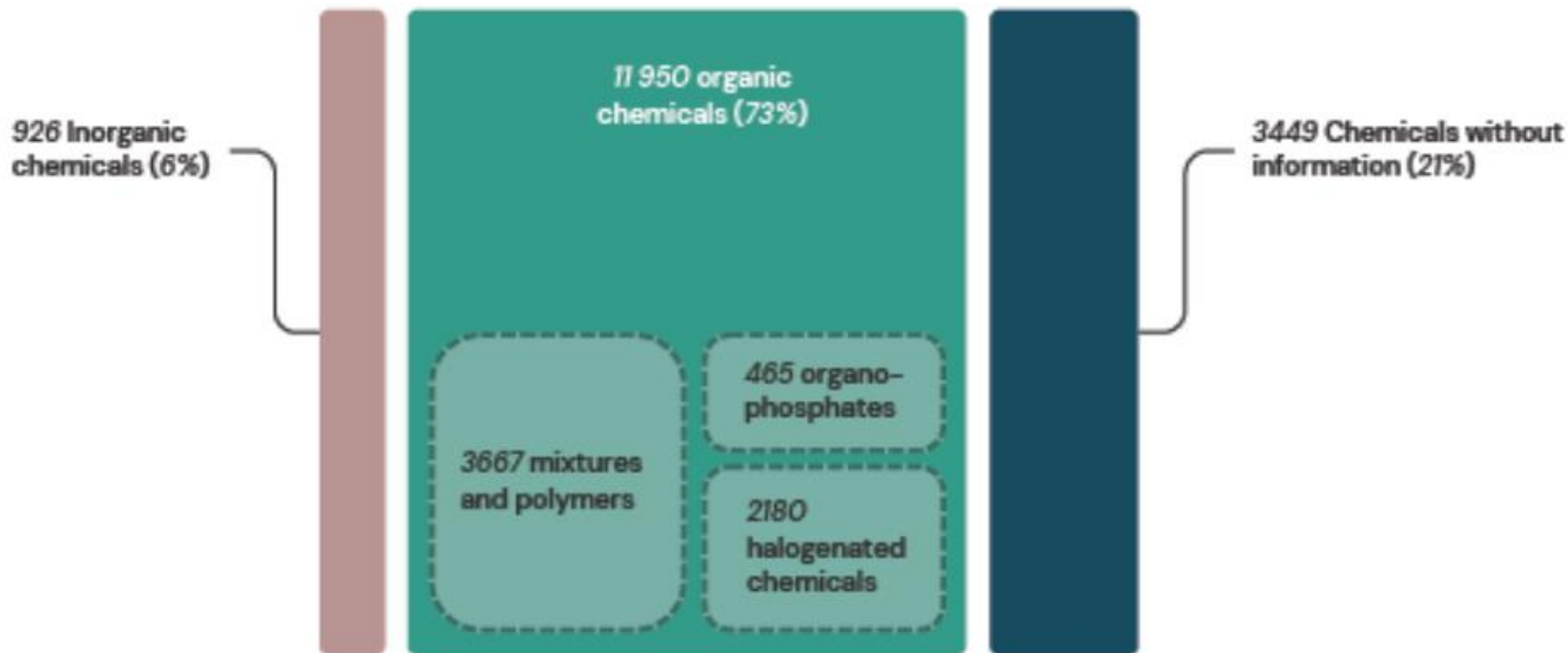
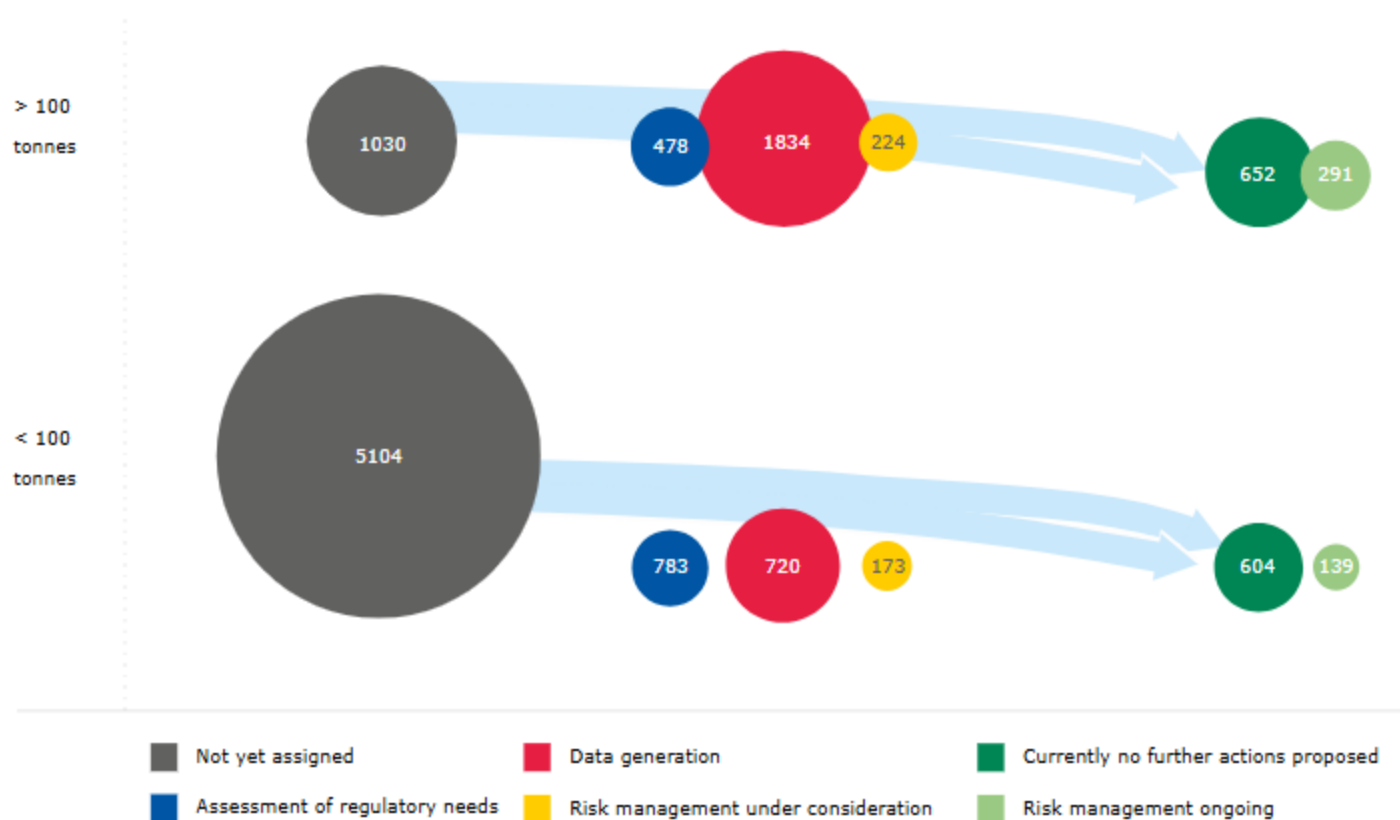


Figure 3: Landscape of plastic chemicals. The mixtures and polymers include 1103 substances of unknown or variable composition, complex reaction products or biological materials (UVCBs).

ECHA's Universe of Registered Substances <https://echa.europa.eu/universe-of-registered-substances>

Figure: REACH chemical universe: substances with active registrations above 1 tonne/year.

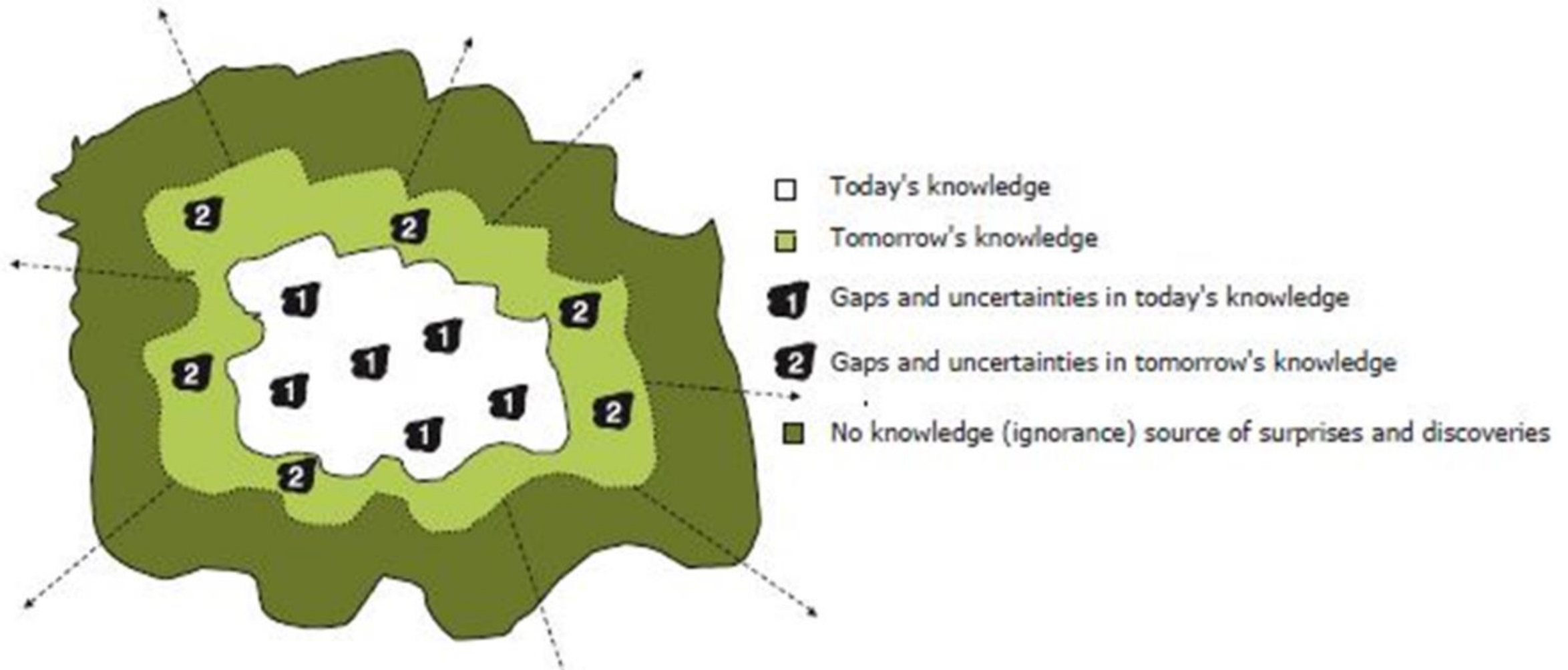


> 100 tonnes: Substances for which there is at least one active registration under Article 10 of REACH registering at a tonnage above 100 tonnes per year.
1 - 100 tonnes: Substances for which there is at least one active registration under Article 10 of REACH registering at a tonnage between 1 and 100 tonnes per year, and for which there are no active registrations registering at a tonnage above 100 tonnes per year under Article 10 of REACH.

Knowledge(s) (and Time)

EEA's Late Lessons from Early Warnings

<https://www.eea.europa.eu/en/analysis/publications/late-lessons-2>



Select the year

1961

2022

Select chemical class

PFAS

Select specific congener

(1mPFOS) 1,1,1,2,3,3,4,4,5,5,6

Select ICD categories

Blood or blood-forming organs

Select study designs

Case study/series, Cross-sectional

Select population

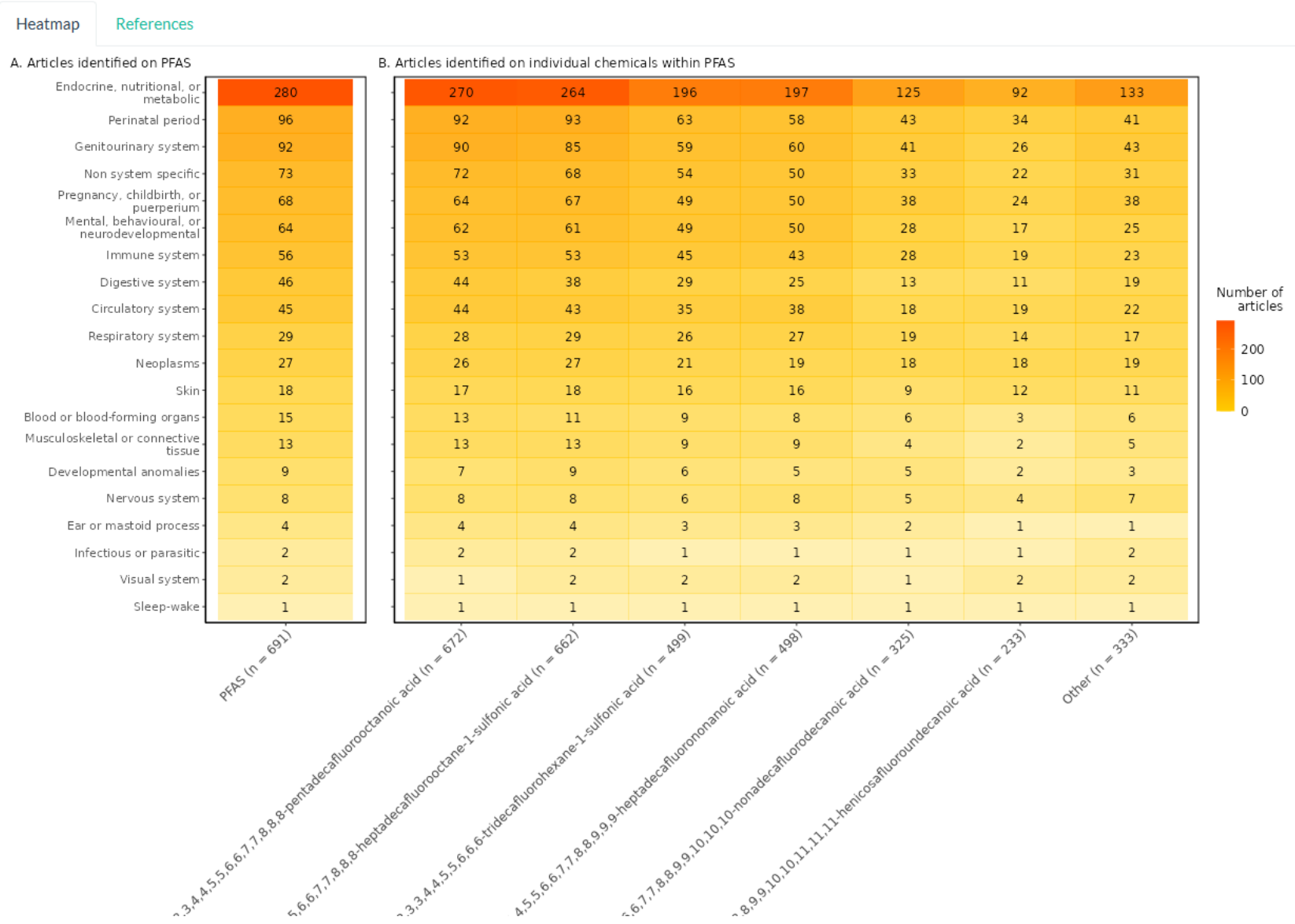
Father-mother-child family, In

Select age exposure group

Prenatal (<0), Neonate (0-1 m

Select age health group

Prenatal (<0), Neonate (0-1 m



Colzin et al (2024) A plausibility database summarizing the level of evidence regarding the hazards induced by the exposome on children health, *International Journal of Hygiene and Environmental Health*, 256: 114311, <https://doi.org/10.1016/j.ijheh.2023.114311>

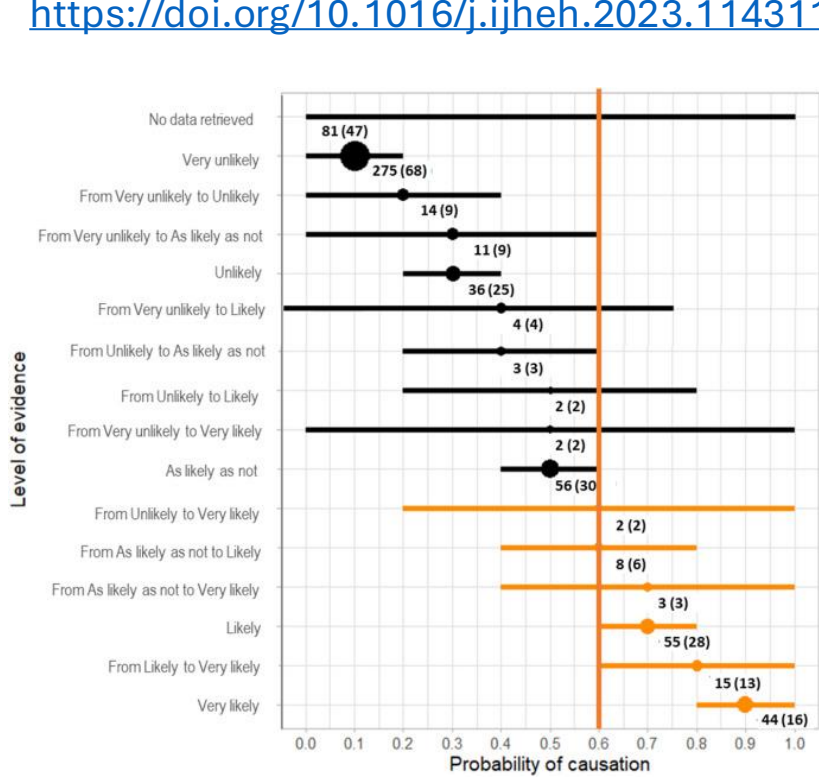


Fig. 2. Distribution of the number of factor-outcome pairs according to the overall LoE and the corresponding uncertainty (indicated by the horizontal line). Numbers indicate the number of factor-outcome pairs (the corresponding number of environmental factors is given in the parentheses). Factor-outcome pairs with a probability of causation above 60% are shown in orange.

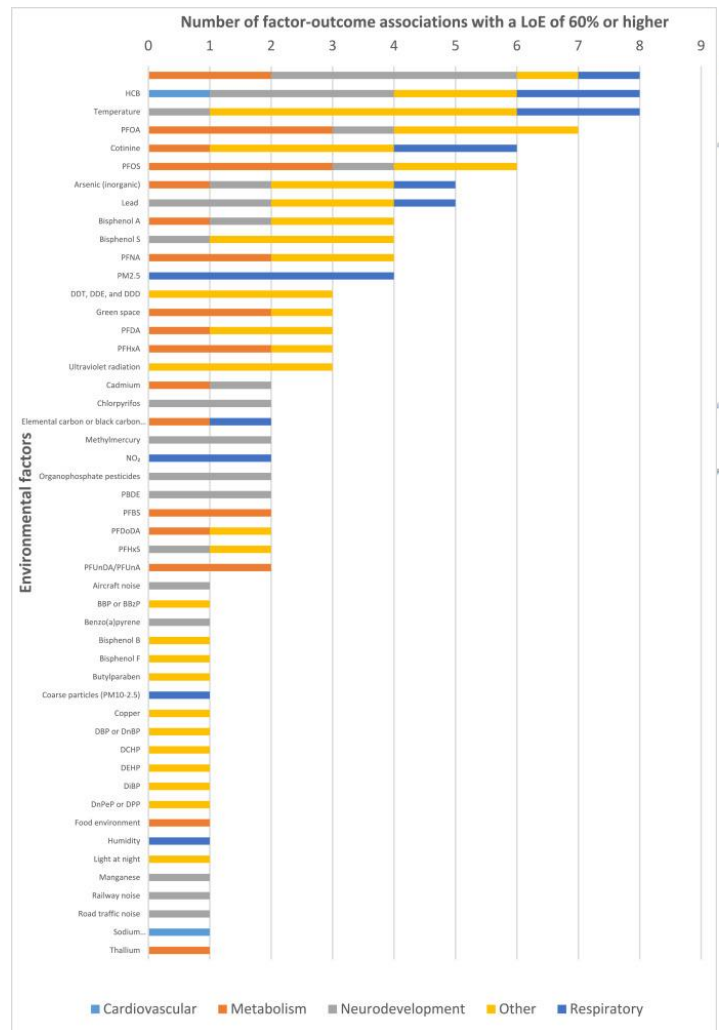


Fig. 3. Number and family of health outcomes among the 127 factor-outcome pairs (corresponding to 49 factors) with an overall LoE of 60 % or more (central estimate of the reliability interval).

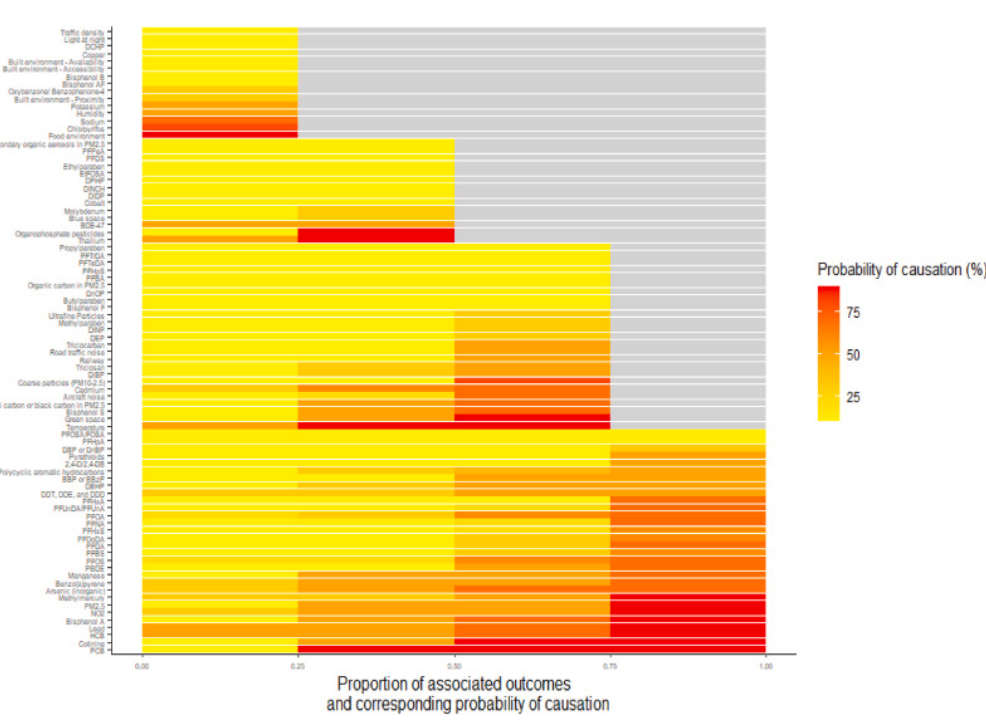
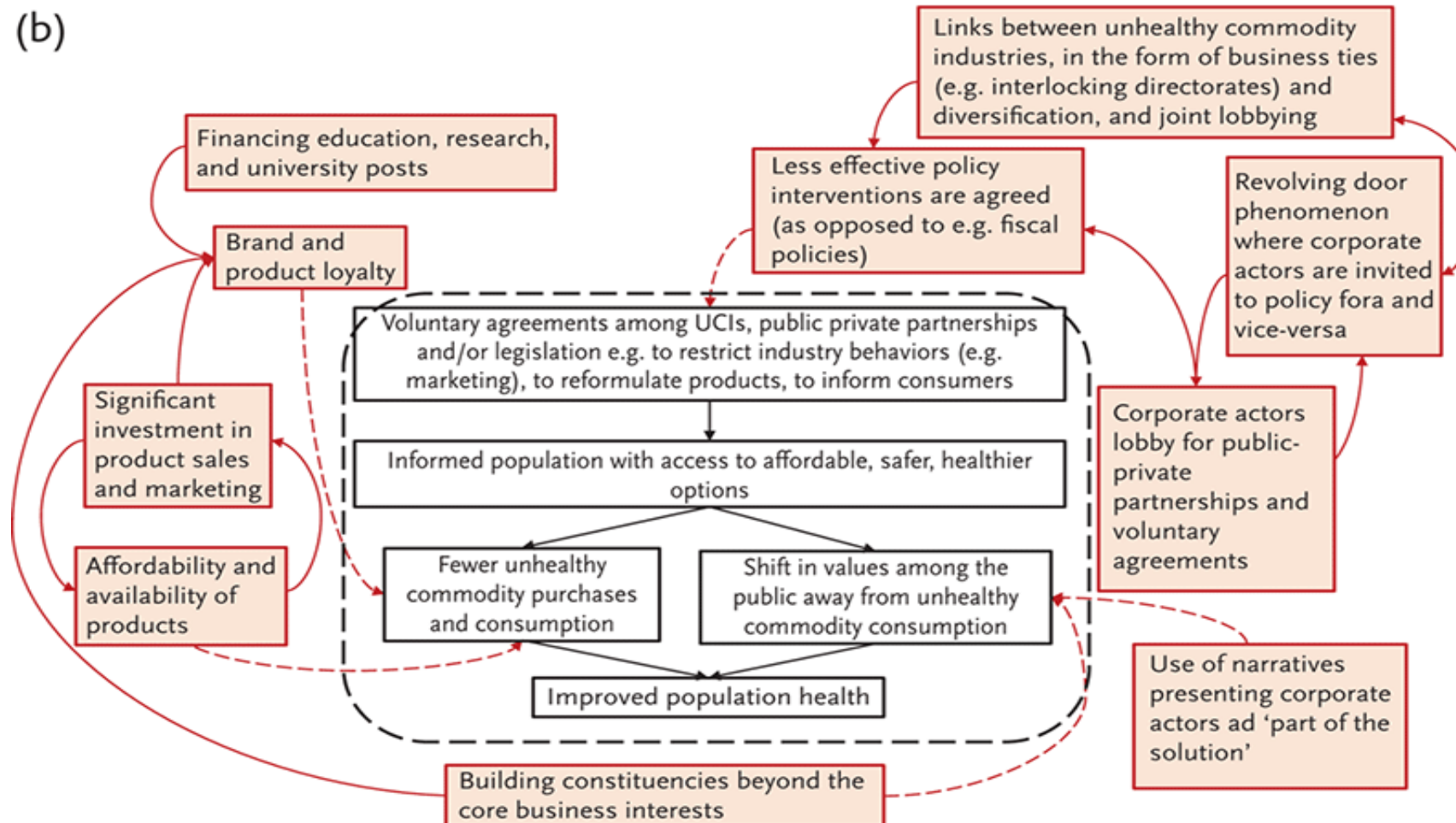


Fig. 4. Distribution of the highest level of evidence by family of health outcome across exposure factors. For simplicity, the graph was limited to effects in the four main axes of cardiovascular, metabolic, neurologic and respiratory health. Substances were sorted by increasing level of evidence. In each of the four health outcome domains, the color indicates the outcome with the highest level of evidence with the considered exposure factor. Grey indicates a lack of data. Exposure factors were ranked by increasing overall LoE across all four health outcomes domains (cardiovascular, metabolism, neurodevelopment, respiratory health).

The Commercial Determinants of Health, <https://doi.org/10.1093/oso/9780197578742.003.0002>

Figure 2.1 C2.F1 Demonstrating differences between linear (a) and systems (b) thinking in approaching commercial determinants of health through the example of voluntary agreements between unhealthy commodity industries (UCIs) and governments.



Gerassimidou et al (2022) Unpacking the complexity of the UK plastic packaging value chain: A stakeholder perspective, *Sustainable Production and Consumption* 30: 657-673, <https://doi.org/10.1016/j.spc.2021.11.005>

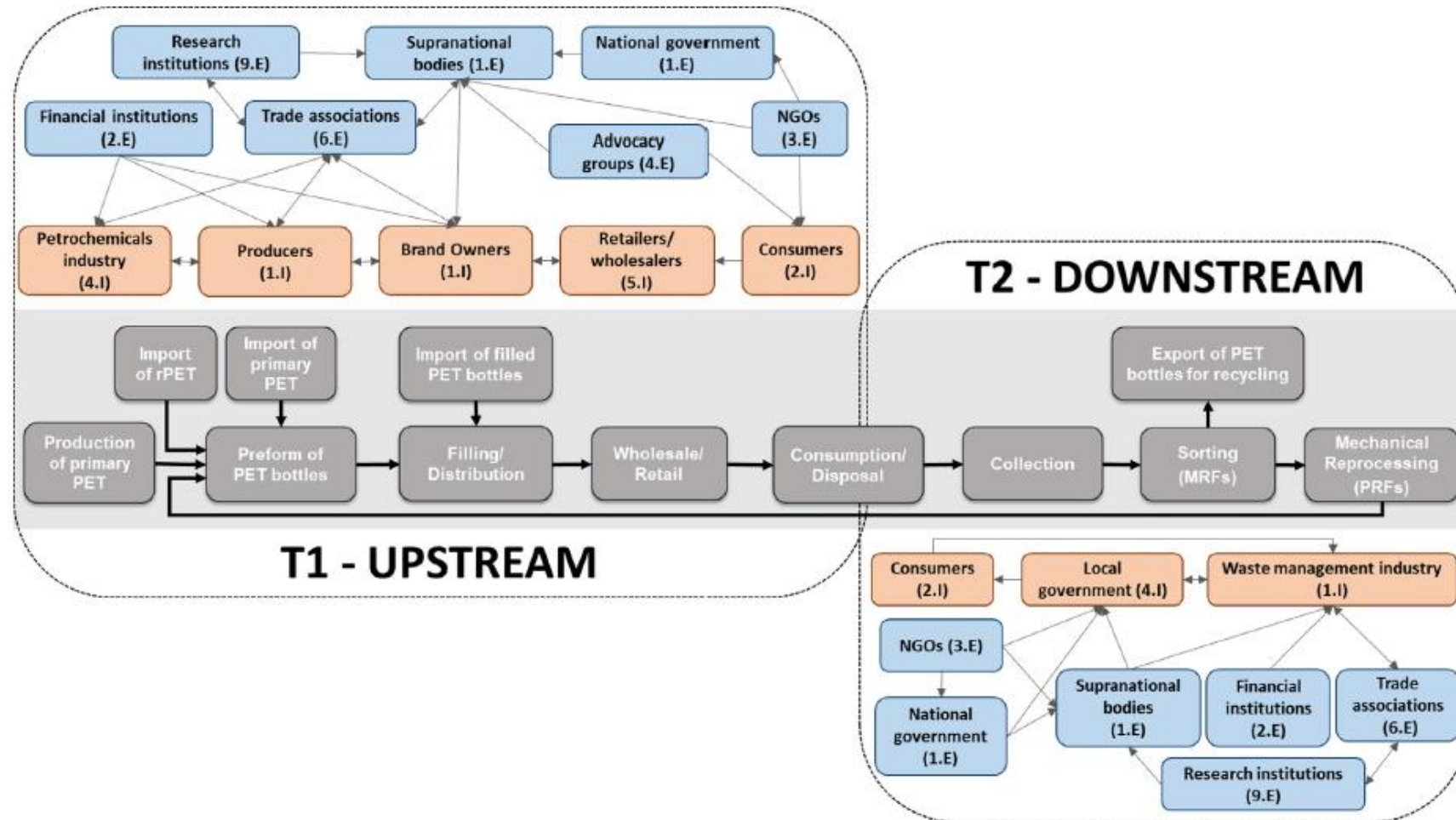


Fig. 3. Depiction of the relationships between the most important internal (n.I) and external (n.E) stakeholders upstream (T1) and downstream (T2) of the PET drinks bottles value chain in the UK.

Sonter, L.J., Dade, M.C., Watson, J.E.M. *et al.* Renewable energy production will exacerbate mining threats to biodiversity. *Nat Commun* **11**, 4174 (2020). <https://doi.org/10.1038/s41467-020-17928-5>

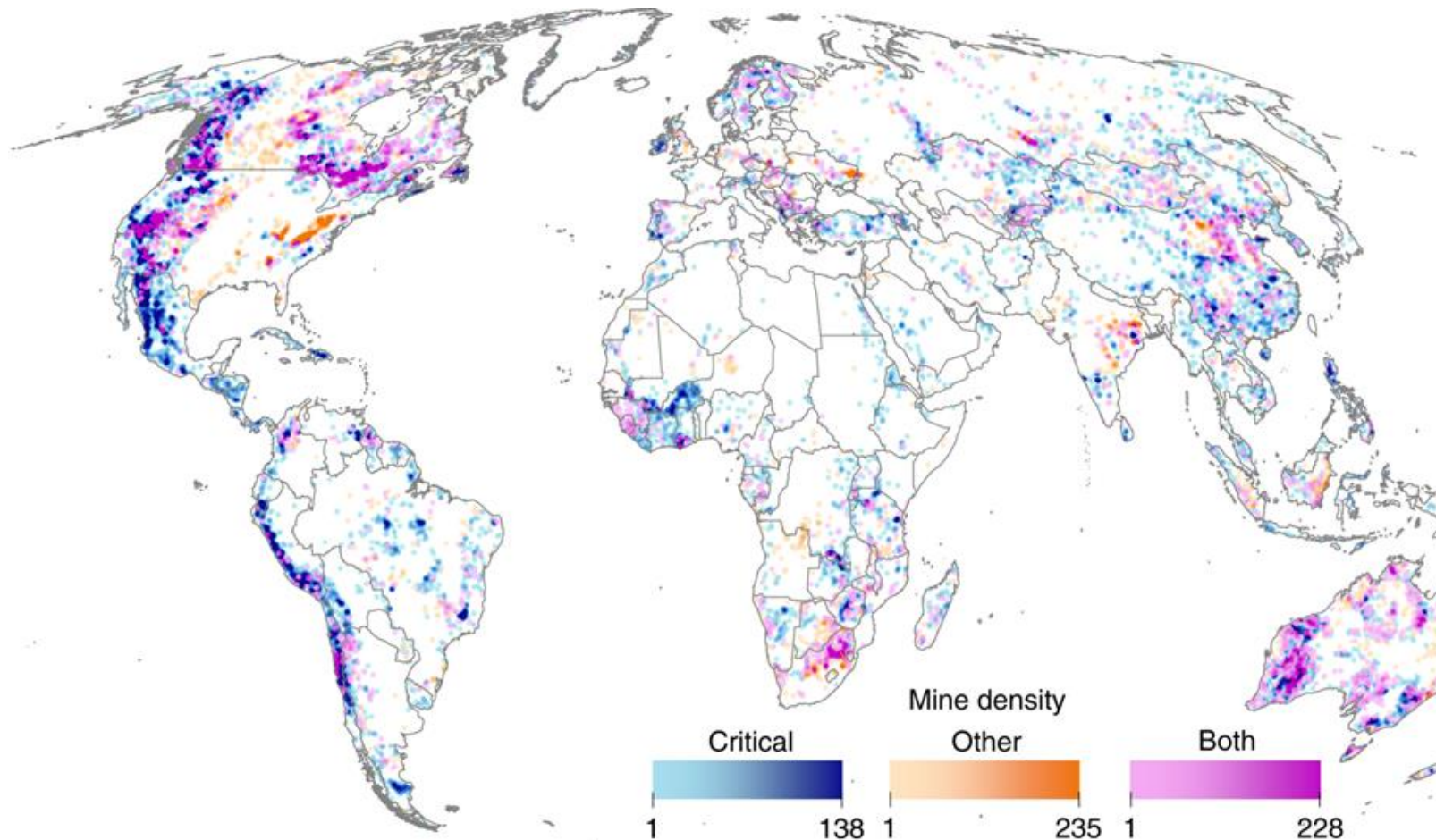


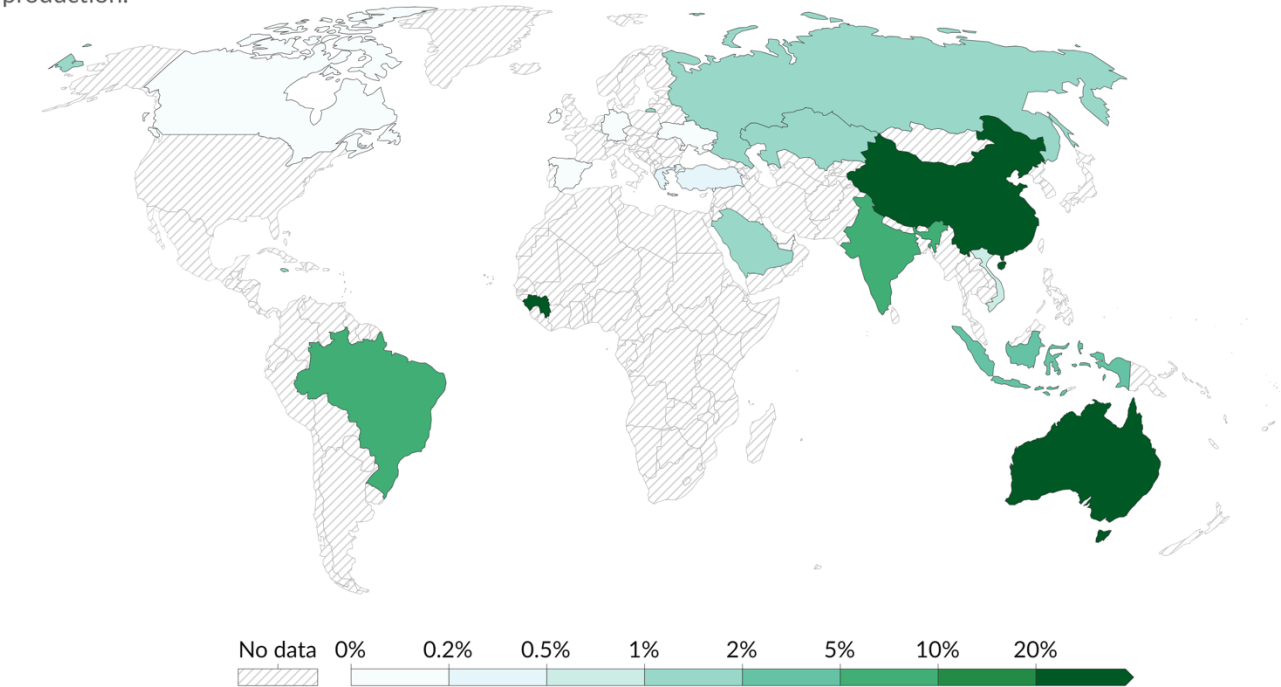
Fig. 1: Global mining areas and their density. Mining areas were mapped using a 50-cell radius around 62,381 pre-operational, operational, and closed mining properties. Mining areas with properties targeting materials critical for renewable energy technology and infrastructure are shown in blue, areas with properties targeting other materials are shown in orange, and those targeting both commodity types are shown in pink. Color shading (light to dark) indicates the density of mining areas—i.e. the number of mining properties within a 50-cell radius of each 1 km cell

Bauxite production as a share of the global total, 2023

Production of bauxite¹ as a share of global bauxite production. Based on mined, rather than refined², production.

Our World
in Data

[Which countries have the critical minerals needed for the energy transition? - Our World in Data](#)



Data source: USGS - Mineral Commodity Summaries (2024); USGS - Historical Statistics for Mineral and Material Commodities (2023); BGS - World Mineral Statistics (2023)

Note: Values are reported as dried bauxite equivalents.

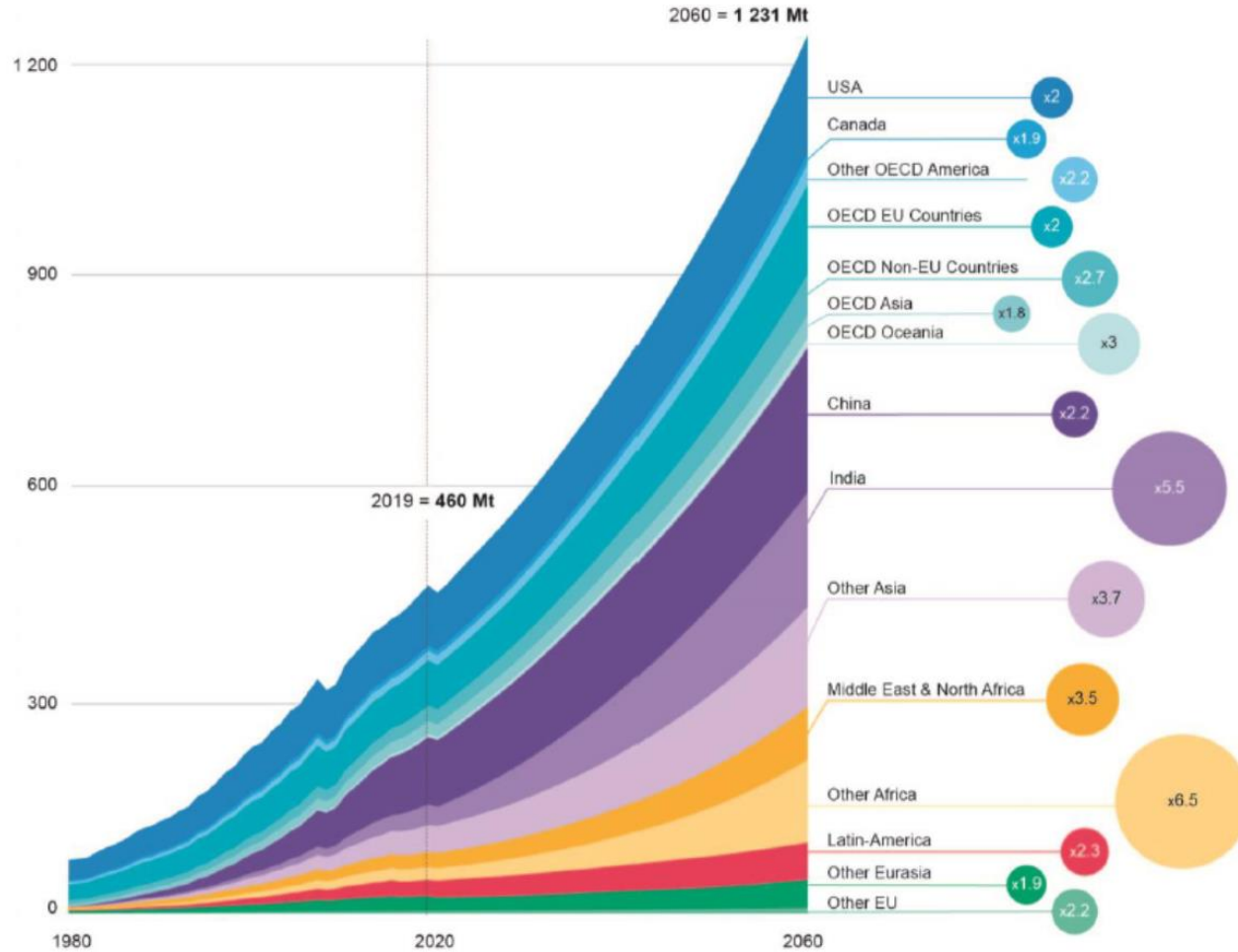
CC BY

1. Bauxite Bauxite is the primary source of aluminum. It's refined into aluminum oxide and then smelted into aluminum. It's essential in construction and for the production of industrial products such as catalysts. It's important for various low-carbon technologies, including wind turbines, solar panels, batteries, electrolyzers, and transmission cables.

2. Refined production Refined production is the process of taking raw materials, such as metals extracted from ores, and purifying them to achieve a higher quality or more usable form. After metals are initially extracted from ore through processes like smelting, they often contain impurities or are not in the ideal form for manufacturing. Refining involves further processing these metals to remove impurities and adjust their composition to meet specific standards. Refined production is essential for transforming raw materials into high-quality metals that are crucial for industries and consumers, from manufacturing electronics to constructing buildings.

Figure 1.3. Plastics use will grow fastest in developing and emerging economies in Africa and Asia

Plastics use in million tonnes (Mt), Baseline scenario



Note: The numbers in the circle on the right-hand side of the graph indicate the growth of plastics use from 2019 (dashed line) to 2060 for each region (e.g. x2 means a doubling of plastics use).

Source: OECD ENV-Linkages model.

OECD Global Plastic Outlook (2022)

https://www.oecd.org/en/publications/global-plastics-outlook_aa1edf33-en.html

Mapping Global Flows of Chemicals: From Fossil Fuel Feedstocks to Chemical Products

Peter G. Levi and Jonathan M. Cullen *Environmental Science & Technology* **2018** 52 (4), 1725-1734

DOI: 10.1021/acs.est.7b04573

Sankey diagram depicting the passage of feedstock through the chemical sector: from fossil fuel feedstocks to chemical products.

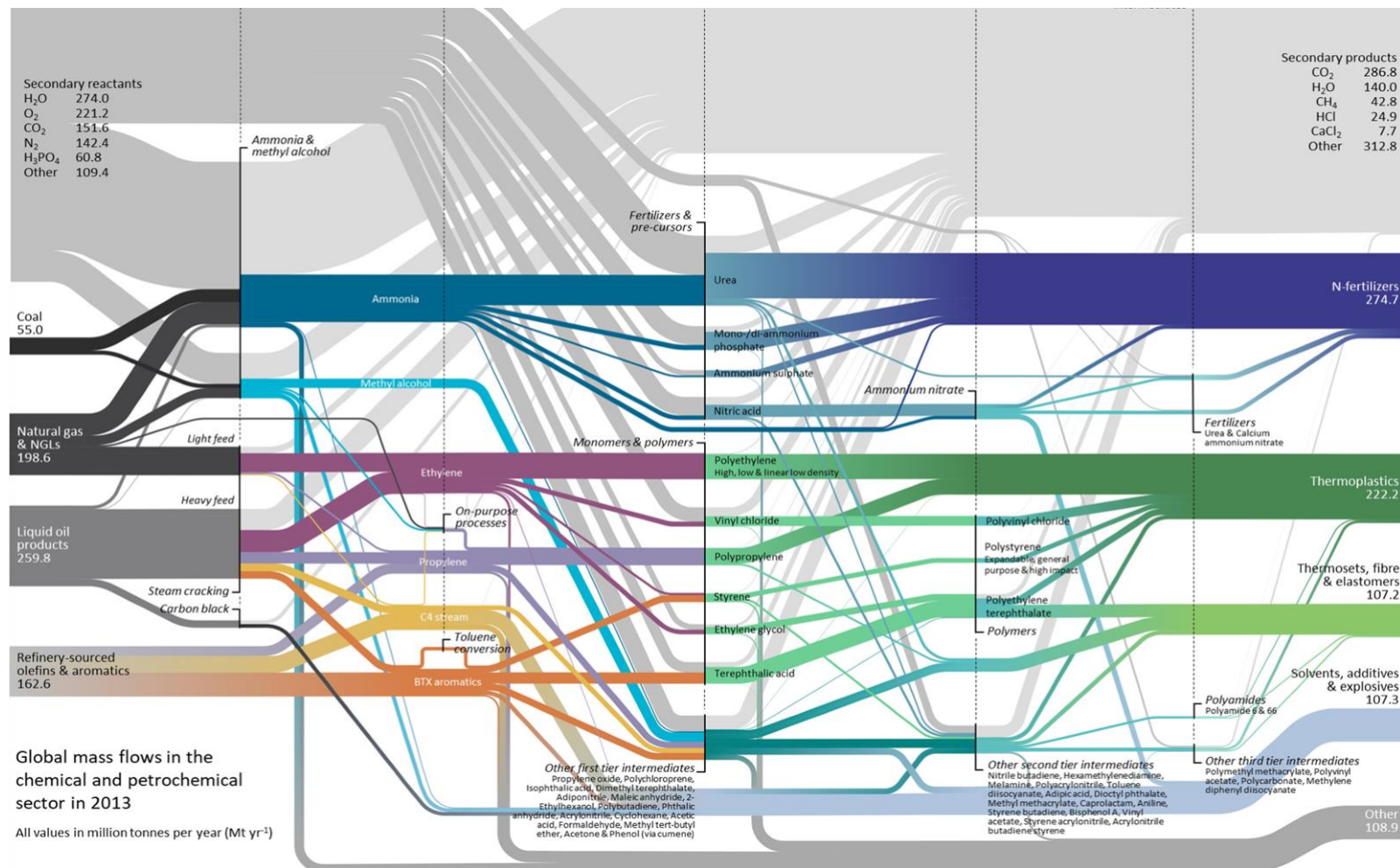
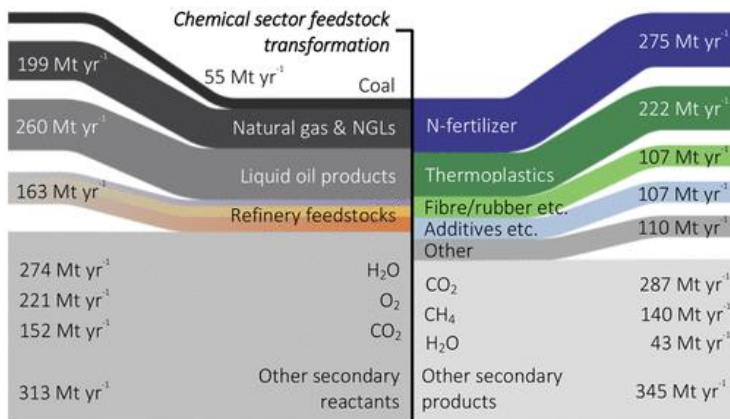
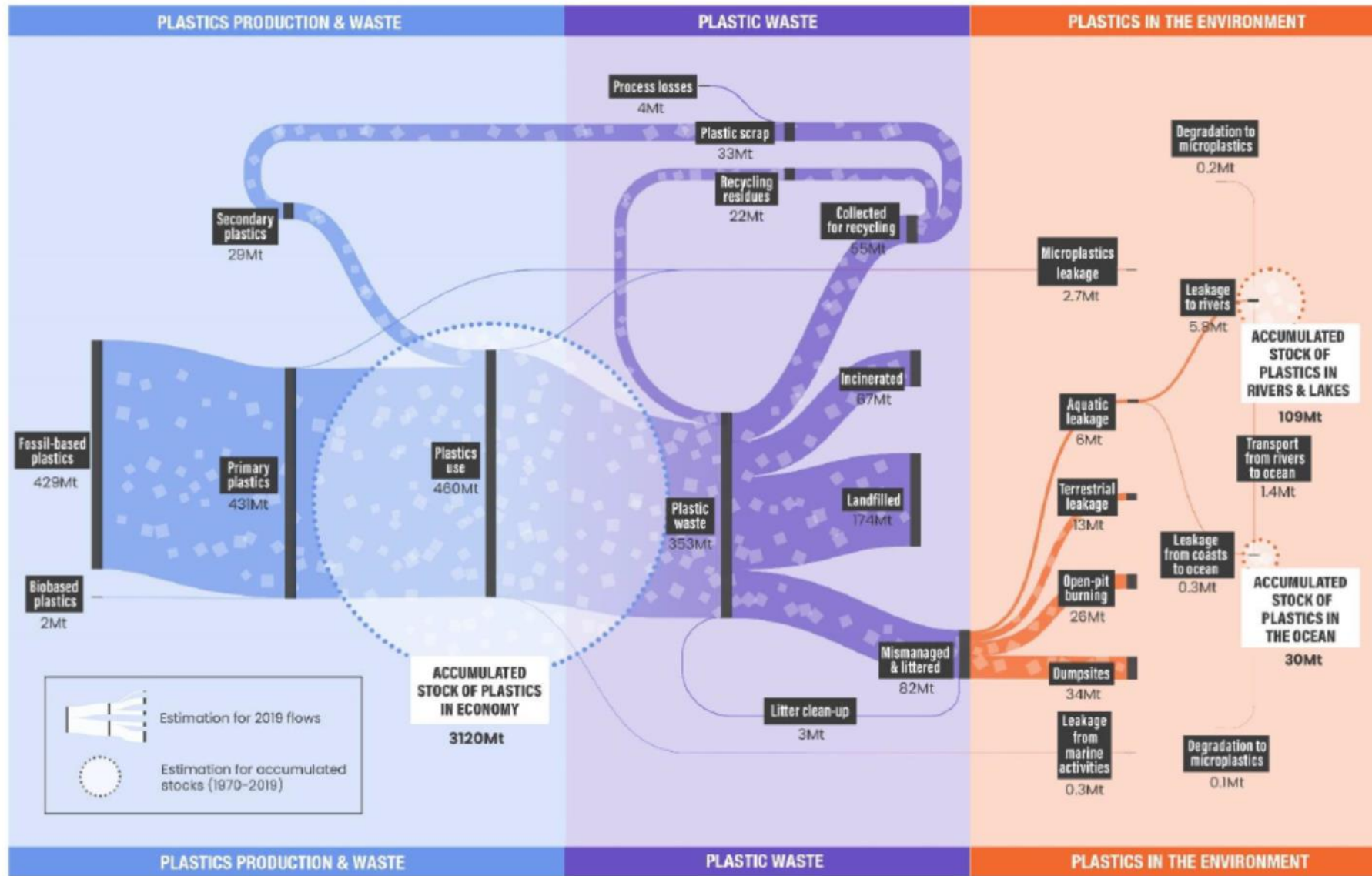


Figure 1.1. Only 33 million tonnes (Mt), or 9% of the 353 Mt of plastic waste, was recycled in 2019

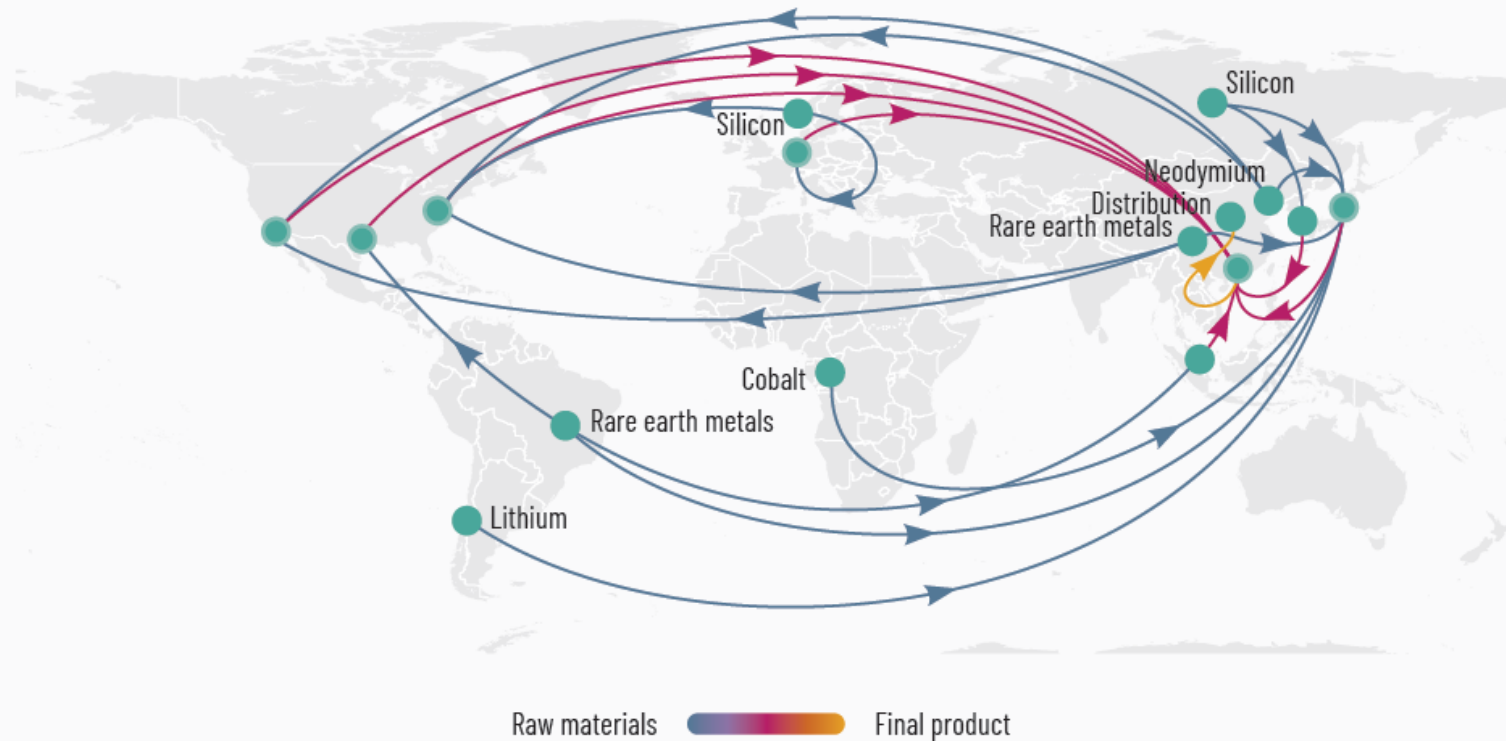


Source: OECD Global Plastics Outlook Database, <https://doi.org/10.1787/c0821f81-en>.

UNEP's Global Chemical Outlook II, 2019

<https://www.unep.org/topics/chemicals-and-pollution-action/chemicals-management/global-chemicals-outlook>

Figure 4.1 Illustration of the complexity of global supply chains: the case of an electronic product (adapted from Sourcemap 2012)



Chemical-intensive products, such as the electronic product depicted here, are traded through increasingly complex global supply chains, spanning many countries and regions. This poses a variety of management challenges.

Panarchy and the adaptive cyc

